

Electromagnetism 2

(spring semester 2026)

Lecture 3

Ideal conductors; electrostatics

- ❖ Ideal conductors in electrostatic field
- ❖ The Laplace equation of electrostatics
- ❖ The uniqueness theorem
- ❖ The method of image charges

Previous lecture

- ❖ Laws of electrostatics and magnetostatics *in free space, in the differential form*:

Gauss law (*universally valid*)

$$\nabla \vec{E} = \rho / \epsilon_0$$

Conservative nature of the **E**-field (*static field only*)

$$\nabla \times \vec{E} = 0$$

Absence of magnetic poles (*universally valid*)

$$\nabla \vec{B} = 0$$

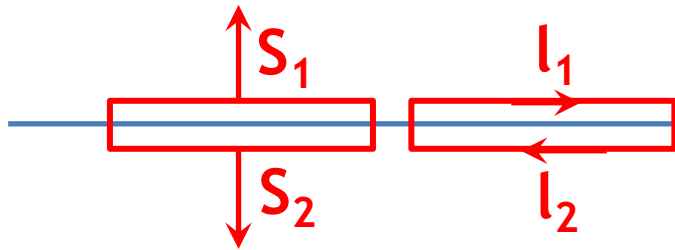
Ampere's law (*static field only*)

$$\nabla \times \vec{B} = \mu_0 \vec{j}$$

- ❖ Electrostatic field is described by a scalar potential, $\vec{E} = -\nabla \varphi$
- ❖ Magnetic field is described by a vector potential, $\vec{B} = \nabla \times \vec{A}$

Boundary conditions

for a thin electrically charged sheet



Surface charge density: σ [unit: C/m]

Gauss law for a very thin (“pillbox”) cylinder S :

$$\int_S \vec{E} d\vec{S} = \frac{q}{\epsilon_0}, \text{ therefore } \vec{E}_1 \vec{S}_1 + \vec{E}_2 \vec{S}_2 = \frac{\sigma S}{\epsilon_0}$$

Using $\vec{S}_1 = -\vec{S}_2$, we obtain for the normal field component:

$$E_{1n} - E_{2n} = \frac{\sigma}{\epsilon_0}$$

For a very thin rectangular loop, $\oint_L \vec{E} d\vec{l} = \vec{E}_1 \vec{l}_1 + \vec{E}_2 \vec{l}_2 = 0$

similarly leads to $E_{1t} = E_{2t}$ for the tangential field component

Conductors in electrostatic field

Conductor: often a lattice of ions and a gas of free electrons. Due to the redistribution of **free charges**, electrostatic field inside **ideal conductor** in the **static case**: $\vec{E} = 0$.

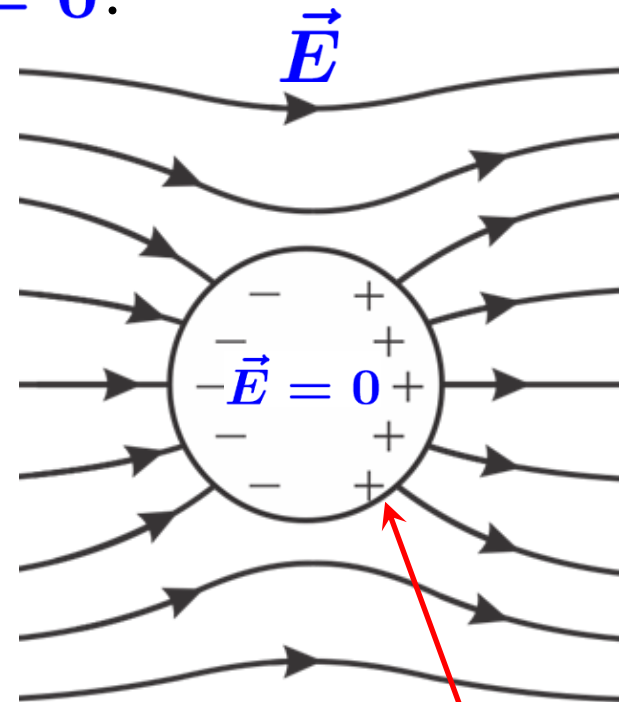
Conductor volume is equipotential: $\phi = \text{const}$

1) Gauss law, any point inside a conductor:

$$\rho = \epsilon_0 \nabla \cdot \vec{E} = \epsilon_0 \nabla \cdot (\vec{0}) = 0$$

2) Conductor in an external static **E** field. It follows from the boundary conditions that the **field is perpendicular to the surface**:

$$E_n = \frac{\sigma}{\epsilon_0} \quad \text{and} \quad E_t = 0$$



Induced surface
electric charges
(unknown a priori)

There are no volume charges inside a conductor ($\rho = 0$).
All charges are localised at the surface ($\sigma \neq 0$ in general).

The Laplace equation

Equations of electrostatics: $\nabla \vec{E} = \rho / \epsilon_0 ; \nabla \times \vec{E} = 0$

It is possible to define the electrostatic potential, $\vec{E} = -\nabla \varphi$
considering that $\nabla \times (\nabla \varphi) = 0$ (lecture 2)

Electrostatic potential at a point A:
(integral along *any* line) $\varphi_A = \int_A^\infty \vec{E} d\vec{l}$

Poisson equation in free space: $\nabla (\nabla \varphi) = \nabla^2 \varphi = -\rho / \epsilon_0$

Widely used in physics: description of heat flow, diffusion, ...

In the absence of free charges, *Laplace equation*: $\nabla^2 \varphi = 0$
(and polarisation charges, lectures 7-8)

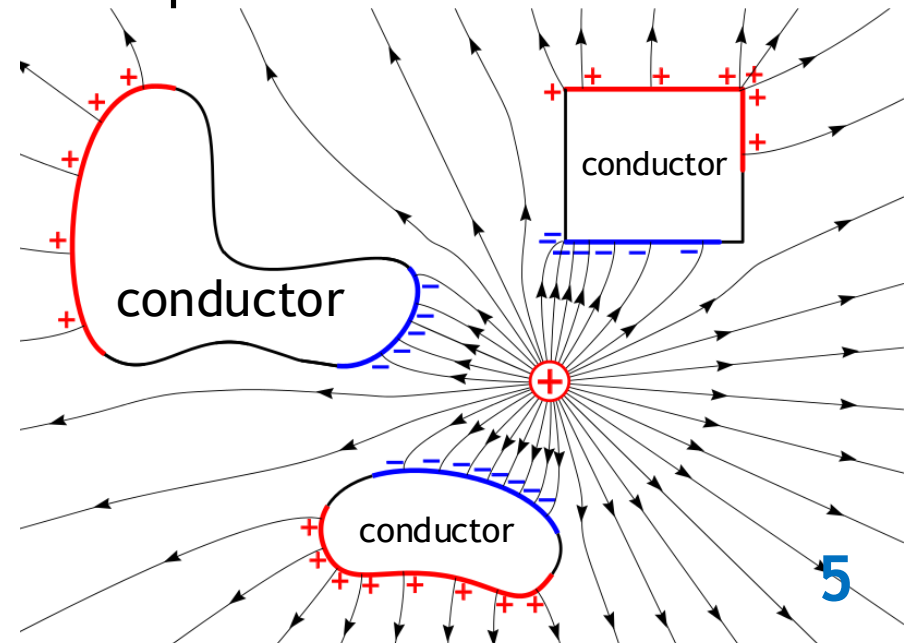
Explicitly in the
Cartesian coordinates, $\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0$

Electrostatic computations

- ❖ Field computation for a *known distribution of charges*:
Coulomb law and the principle of superposition (*lecture 1*).
- ❖ But distribution of *free charges* in conductors is *not known* a priori.

The **general problem of electrostatics** (in free space):

- ❖ Several conducting bodies are placed in vacuum.
- ❖ *Boundary conditions*: for each conductor, either its potential ϕ_i or its total charge Q_i is specified.
- ❖ It is required to compute the electrostatic potential $\phi(\mathbf{x}, \mathbf{y}, \mathbf{z})$ in all points of space.
- ❖ After $\phi(\mathbf{x}, \mathbf{y}, \mathbf{z})$ is found, the electric field is computed as $\vec{E} = -\nabla\phi$
- ❖ Densities of the induced charge at the conductor surfaces are found as $\sigma(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \epsilon_0 E_n(\mathbf{x}, \mathbf{y}, \mathbf{z})$, where $E_n(\mathbf{x}, \mathbf{y}, \mathbf{z})$ is the field just outside the conductor.



The uniqueness theorem

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0$$

A solution of Laplace equation does not have local maxima/minima (a local minimum, for example, means $\frac{\partial^2 \varphi}{\partial x^2} > 0$, $\frac{\partial^2 \varphi}{\partial y^2} > 0$, $\frac{\partial^2 \varphi}{\partial z^2} > 0$).

Consider two different solutions for the same set of boundary conditions:

$$\nabla^2 \varphi_1 = 0, \quad \nabla^2 \varphi_2 = 0 ; \text{ therefore } \nabla^2(\varphi_0) = \nabla^2(\varphi_1 - \varphi_2) = 0$$

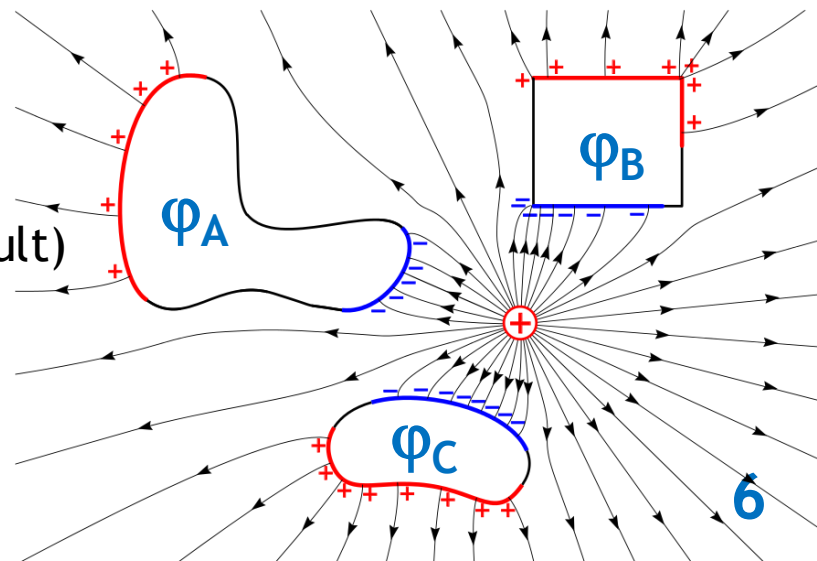
The solution $\varphi_0(\vec{r})$ has no minima/maxima;

the boundary conditions are $\varphi_0 = 0$

$$\text{Therefore } \varphi_0(\vec{r}) \equiv 0$$

(Proof for boundary conditions for $\mathbf{Q}_i$ is more difficult)

Poisson and Laplace equations have a unique solution for any complete set of boundary conditions.



Faraday cage

Independently of the method chosen, if a solution to $\nabla^2 \varphi = 0$ is found (or guessed), it is *the only solution*.

For a cavity in a conducting body: inside the cavity, $\nabla^2 \varphi = 0$

On the surface of the cavity, $\varphi = \varphi_0 = \text{const}$

A (unique!) solution inside the cavity: $\varphi(\vec{r}) \equiv \varphi_0$

Therefore, no electric field in the cavity:

$$\vec{E} = \nabla \varphi_0 = \vec{0}$$

❖ “Faraday cages” are used for electrostatic shielding of equipment, in USB/coaxial cables, in forensics.

More generally, fields inside and outside a conductive enclosure are independent.



The method of image charges

For a system of charges and a grounded ($\varphi=0$) conductor:

- ❖ there are *induced charges* on the conductor surface;
- ❖ using the uniqueness theorem, the conductor (and induced charges) can be replaced by a system of *image charges* in such a way that the conductor surface remains at $\varphi=0$.

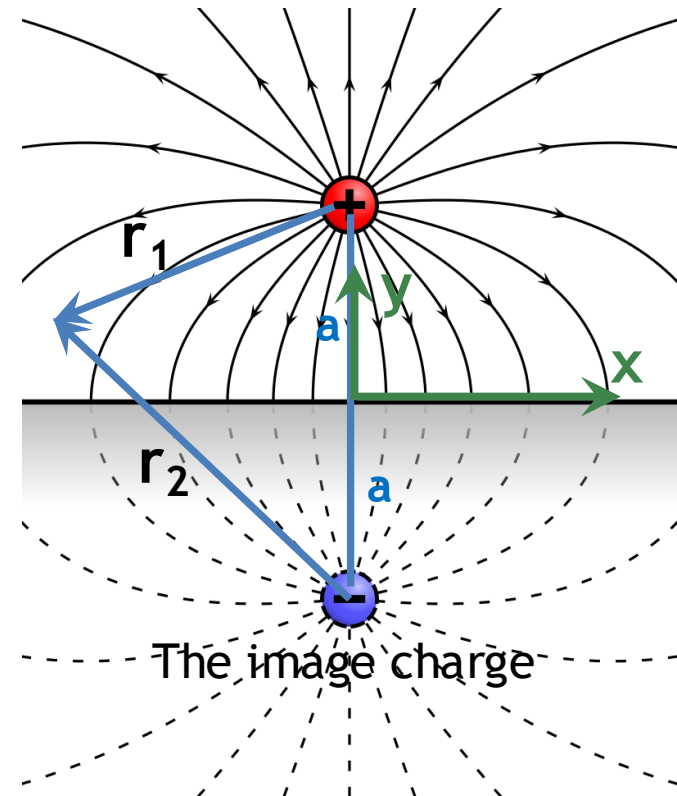
Example: a point charge Q at a distance a from an infinite conducting plate of any thickness.

Consider an *image charge* $-Q$ placed symmetrically at a distance a below the surface.

Potential due to the real and image charges:

$$\varphi(\vec{r}) = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

At the surface, $\varphi|_{y=0} = 0$:
the image charge is equivalent to the plate.



Example (continued)

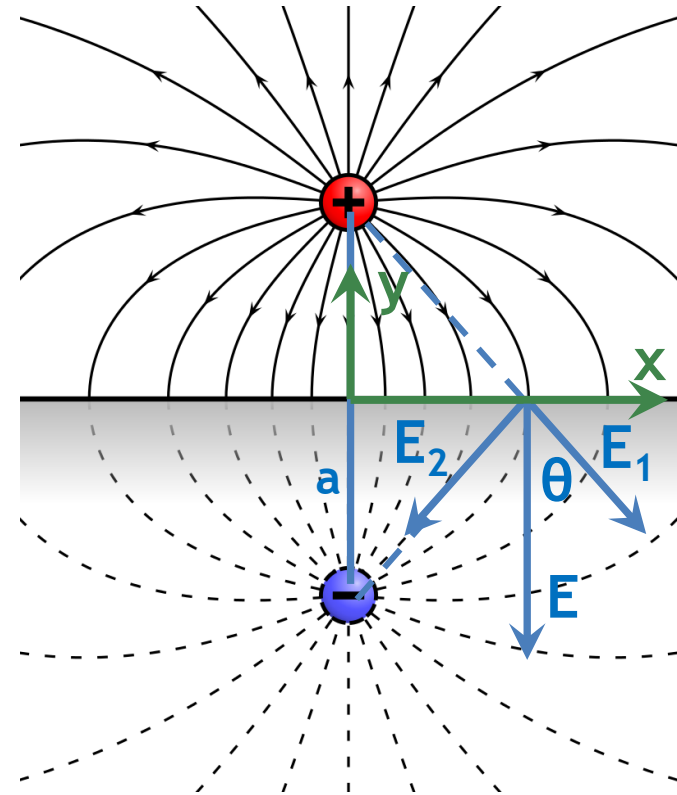
Above the surface, the field of the *induced charges* is equivalent to the field of the *image charge*.

Attractive force between point charge and the plate:

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{(2a)^2}$$

Density of the induced surface charge, obtained using Coulomb's law:

$$\begin{aligned}\sigma(x) &= \epsilon_0 E_y(x) = -2\epsilon_0 E_1 \cos \theta = \\ &= -\frac{2\epsilon_0}{4\pi\epsilon_0} \cdot \frac{Q}{x^2 + a^2} \cdot \frac{a}{\sqrt{x^2 + a^2}} = \\ &= -\frac{Q}{2\pi a^2} \left(\frac{a}{\sqrt{x^2 + a^2}} \right)^3 = -\frac{Q}{2\pi a^2} \cos^3 \theta\end{aligned}$$



Total charge induced

Total (negative) charge induced on the surface:

$$\begin{aligned} Q_{\text{ind}} &= \int_0^{\infty} \sigma(x) \cdot 2\pi x dx = -\frac{Q}{2\pi a^2} \int_0^{\infty} 2\pi x \left(\frac{a}{\sqrt{x^2 + a^2}} \right)^3 dx \\ &= -\frac{Qa}{2} \int_0^{\infty} \frac{d(x^2)}{(x^2 + a^2)^{3/2}} = -\frac{Qa}{2} \int_{a^2}^{\infty} \frac{dt}{t^{3/2}} = \frac{Qa}{2} \cdot \frac{2}{\sqrt{t}} \Big|_{a^2}^{\infty} = -Q \end{aligned}$$

We can also see that $Q_{\text{ind}} = -Q$ using the Gauss law:

for a very large sphere, surface area $A \sim r^2$;

dipole field $E \sim 1/r^3$ (lecture 4); electric field flux through the surface:

$$\int_S \vec{E} d\vec{S} \sim \frac{1}{r} \rightarrow 0$$

Therefore, the net charge enclosed is $Q_{\text{ind}} + Q = 0$

Magnetostatics

[not discussed in the lecture]

Equations of magnetostatics (steady currents, $\mathbf{j}=\text{const}$):

$$\nabla \vec{B} = 0 \quad \text{and} \quad \nabla \times \vec{B} = \mu_0 \vec{j}$$

Considering that $\nabla(\nabla \times \vec{A}) = 0$ for any vector field, one can define the *vector potential*, $\vec{A}$, as follows:

$$\vec{B} = \nabla \times \vec{A}; \quad \nabla \vec{A} = 0 \quad \text{“Coulomb gauge”}$$

Substituting into Ampere's law,

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \vec{A}) - \nabla^2 \vec{A} = -\nabla^2 \vec{A} = \mu_0 \vec{j}$$

Finally, $\boxed{\nabla^2 \vec{A} = -\mu_0 \vec{j}}$

Equivalently, $\nabla^2 A_x = -\mu_0 j_x; \quad \nabla^2 A_y = -\mu_0 j_y; \quad \nabla^2 A_z = -\mu_0 j_z$

Analogous to the Poisson eqn of electrostatics: $\nabla^2 \varphi = -\rho/\epsilon_0$

This provides a general method for magnetostatic problems.

- ❖ Ideal conductors in electrostatic field:
 - ✓ the volume of the conductor is equipotential;
 - ✓ there are no volume charges ($\rho=0$);
 - ✓ in general, there are non-zero surface charges ($\sigma \neq 0$);
 - ✓ just above the surface, $\mathbf{E}_n = \sigma/\epsilon_0$ and $\mathbf{E}_t = 0$.
- ❖ Laplace equation for the electrostatic potential in the absence of free charges:

$$\nabla^2 \varphi = 0$$

- ❖ The uniqueness theorem: the Laplace equation has a unique solution for any complete set of boundary conditions.
- ❖ The method of image charges: a tool for electrostatic field computations in the presence of conductors.